

CASE 1

The Mystery of the Upside-Down Map



DETECTIVE:

Five thinking missions. Five story unlocks. One very suspicious duck.

Your path matters more than being fast.

How to Play

- 1 Read the story until you reach a locked Thinking Mission.
- 2 Build a path. Draw, write, test, cross things out - thinking can look messy.
- 3 If you get stuck, visit the Hint Station and uncover one hint at a time.
- 4 Finish the mission, then turn the page to unlock what happens next.
- 5 At the end, tally the thinking badges you used during the case.

THINKING BADGES

**Clue Finder**

I noticed something useful.

**Path Builder**

I made a way to start.

**Brave Try**

I tried before I was sure.

**Check-It**

I checked if it made sense.

**Try Again**

I changed course and kept going.

REMEMBER: You do not have to know right away. You just need a next useful step.

A Very Suspicious Package



Remy the Raccoon was having a serious detective morning. Which meant he was wearing his detective hat, his detective backpack, his detective magnifying glass... and one fuzzy purple slipper.

Then - THUMP! A small yellow package landed outside the Thinking Detective Clubhouse. It had no name and no return address. Only this message: FOR A THINKING DETECTIVE WHO IS NOT AFRAID TO GET CONFUSED.

Inside was an old folded map and a note: Find the five clues. And whatever you do... DO NOT TRUST THE DUCK. Across the street stood a duck. Remy stared. The duck stared back. "Suspicious," he whispered.

The map seemed completely wrong. Remy scratched his head. "I officially have no idea what's going on. Excellent. That's when the thinking starts."

Complete Mission 1 to discover the first real clue.

Which Way Is Up?

PRIMARY

Representation & Model Building

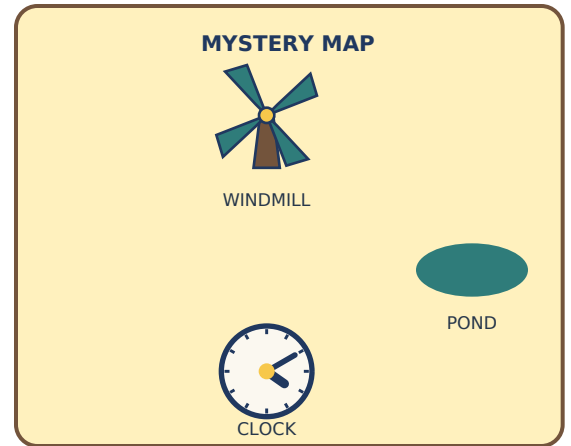
SECONDARY

Anchor Finding / Checking

Remy stands on the clubhouse porch. The clock tower is straight ahead, the pond is on his left, and the windmill is behind him.

But on the map, the clock is at the bottom, the pond is on the right, and the windmill is at the top.

What could Remy do to the map so it matches what he sees around him? You do not need to redraw anything.



You may turn this map card - or the whole page.

SHOW YOUR PATH FIRST

YOUR ANSWER

BEFORE YOU TURN THE PAGE: Can you check your idea using all three landmarks?

Hint Station

Only uncover one hint at a time. Cover the lower boxes with another sheet of paper and slide it down only when you want more help.

1

Pick just one landmark. If the clock tower is straight ahead, where should it be on the map?

2

Instead of moving the landmarks, could you move the paper?

3

Try turning the map around. Then check the pond and windmill too.

REMEMBER: A hint is not a failure. It is another clue when your brain needs a foothold.

The Clock With No Hands



Remy tried his idea on the map. Suddenly, tiny writing along the edge made sense: **FIRST CLUE: WHERE TIME STANDS STILL.**

He remembered the old clock in the town square. It had not worked in years. Remy raced over. The duck followed. "Stop following me, suspicious duck!" QUACK.

At the base of the clock, a plaque said it had stopped at 4:10. Another line said the first Thinking Detective hid a blue envelope 35 minutes before the clock stopped.

Remy looked up at the frozen clock face. "Okay," he said. "Time to think backward."

Solve Mission 2 to find the time that opens the clock compartment.

The Clock That Stopped

SECONDARY Anchor Finding / Decomposition

The clock stopped at 4:10.
The blue envelope was hidden 35 minutes
before the clock stopped.

A horizontal timeline with a dark blue line. On the right end, there is a vertical tick mark labeled "4:10". On the left end, there is a vertical tick mark with an orange triangle pointing left towards it, and a red question mark "?" above the triangle. In the center of the timeline, there is a vertical tick mark labeled "useful anchor" in green. Below the timeline, the text "MOVE BACKWARD 35 MINUTES" is written in orange.

Thinking Detectives - Case 1

Hint Station

Only uncover one hint at a time. Cover the lower boxes with another sheet of paper and slide it down only when you want more help.

1 Are you moving forward in time or backward?

2 Is there an easy time close to 4:10 that could be a useful anchor?

3 Go backward from 4:10 to 4:00 first. How many of the 35 minutes have you used?

4 You used 10 minutes. There are 25 more minutes to travel backward.

REMEMBER: A hint is not a failure. It is another clue when your brain needs a foothold.

Three Paths and One Very Bad Shortcut



Remy worked backward through the time clue. At the correct time, he pressed the matching number. CLICK! A secret compartment opened. Inside was a blue envelope.

The message read: THE NEXT CLUE WAITS WHERE THREE SMALL PATHS BECOME ONE.

At the edge of Whispering Woods, three trails joined beside a giant oak. A note told Remy to find Nose Rock, the Y-Tree, and the Squeaky Bridge on the way to a hidden shed.

Remy tried searching for all three at once. Left! Right! Left! Then backward into a bush. FWUMP. The duck watched. "Not one word," Remy said. Quack.

Mission 3 will reveal how much trail remains after the bridge.

The Three-Part Trail

SECONDARY Anchor Finding / Representation

WHOLE JOURNEY = 12 MARKERS

3 4 2 ?

OAK NOSE ROCK Y-TREE BRIDGE SHED

How many more trail markers are there from the Squeaky Bridge to the hidden shed? Build or draw the whole trail if that helps.

SHOW YOUR PATH FIRST

YOUR ANSWER - FROM THE BRIDGE TO THE SHED

Hint Station

Only uncover one hint at a time. Cover the lower boxes with another sheet of paper and slide it down only when you want more help.

1 Which pieces of the 12-marker journey do you already know?

2 Try drawing the whole trail as several connected pieces.

3 How many markers are accounted for by the time Remy reaches the bridge?

4 $3 + 4 + 2 = 9$ markers are accounted for. The whole trail is 12.

REMEMBER: A hint is not a failure. It is another clue when your brain needs a foothold.

Somebody Has Been Here



Remy figured out how much trail remained and followed the markers. Behind some vines stood an abandoned little shed. The door was already open.

"Ohhhhhh," Remy whispered. "Somebody has been here." Inside were three strange objects: a red feather, a muddy boot print, and half of a peanut-butter sandwich.

Three animals had been nearby: Professor Pigeon, Benny Beaver, and Dottie Duck. Remy slowly turned toward Dottie. "I KNEW IT!" QUACK.

Then Remy stopped. The clues did not fit his guess. "Maybe I should figure out what cannot be true before I decide what is true."

Use Mission 4 to discover who was carrying the missing map.

MISSION 4

The Suspicious Shed

PRIMARY Elimination & Constraint Reasoning

SECONDARY Relationship Reasoning / Checking

FACTS

1. Professor Pigeon left the red feather.
2. Benny Beaver hates peanut butter.
3. Dottie Duck does not wear boots.
4. The animal who left the boot print carried the missing map.



FEATHER



BOOT



SANDWICH

FEATHER

BOOT

SANDWICH

PIGEON

BEAVER

DUCK

Who was carrying the missing map? Do not guess. Cross out what cannot be true, then show your path. Check that every animal has exactly one object.

SHOW YOUR PATH FIRST

YOUR ANSWER - THE MAP WAS CARRIED BY

Hint Station

Only uncover one hint at a time. Cover the lower boxes with another sheet of paper and slide it down only when you want more help.

1 Start with the strongest fact. Who definitely left the feather?

2 Once Pigeon has the feather, only the sandwich and boot print remain for Benny and Dottie.

3 Benny cannot have the sandwich. Which object must belong to Benny?

REMEMBER: A hint is not a failure. It is another clue when your brain needs a foothold.

The Duck Knows Something



Remy checked every clue again. Then he gasped. "Dottie... you did NOT take the map! You've been trying to help me?" QUACK QUACK!

"Why didn't you just say so?" Remy asked. Dottie stared at him. Remy paused. "Oh. Right."

Dottie pecked the map. Remy turned it the useful way again. Suddenly the giant red X pointed to a hidden door in the oak tree.

But the door would close at sunset. Remy needed both a brass key and a secret code first. He looked at the sky. "Uh-oh. Time to make a plan."

Mission 5 decides whether Remy can open the door before 6:00.

MISSION 5

Beat the Sunset

PRIMARY Sequencing & Planning

SECONDARY Constraint Reasoning / Checking

It is 5:20 p.m. The hidden entrance closes at 6:00 p.m. Remy needs both the brass key and the secret code before he can open the oak door.

PLAN A - KEY FIRST

- | | | |
|---|-------------------|-------|
| 1 | Shed to Bridge | 5 min |
| 2 | Get key | 5 min |
| 3 | Bridge to Sundial | 8 min |
| 4 | Read code | 5 min |
| 5 | Sundial to Oak | 7 min |
| 6 | Open door | 5 min |

PLAN B - CODE FIRST

- | | | |
|---|-------------------|-------|
| 1 | Shed to Sundial | 7 min |
| 2 | Read code | 5 min |
| 3 | Sundial to Bridge | 8 min |
| 4 | Get key | 5 min |
| 5 | Bridge to Oak | 6 min |
| 6 | Open door | 5 min |

Can Remy make it before 6:00? Build a plan and show your path.
BONUS: Is there more than one plan that works?

SHOW YOUR PATH FIRST

YOUR ANSWER - REMY CAN / CANNOT MAKE IT BECAUSE _____

Hint Station

Only uncover one hint at a time. Cover the lower boxes with another sheet of paper and slide it down only when you want more help.

1 How many minutes are available from 5:20 to 6:00?

2 Pick one plan first. Add every step, including opening the door.

3 Compare your total time with the 40 minutes available.

4 After checking one plan, try the other. Two different paths can both work.

REMEMBER: A hint is not a failure. It is another clue when your brain needs a foothold.

The First Thinking Detective



Remy completed the final plan.

CLICK.

Part of the giant oak swung open. Behind it was a tiny hidden room. Dust floated through golden light. Old detective notebooks covered the shelves. Magnifying glasses hung from hooks.

In the middle sat a small wooden chest. Inside was no gold. No jewels. Only a very old badge: THINKING DETECTIVE #1.

A Thinking Detective isn't someone who always knows what to do.

A Thinking Detective is someone who knows what to do when they DON'T know what to do.

Find something you know. Try something. Check what happened. And keep thinking.

Dottie found Remy's missing purple slipper behind the chest. "How did THAT get here?" Remy yelled. Dottie backed away. QUACK QUACK QUACK! CASE CLOSED? ...Maybe.

Your Thinking Detective Badge Tally

These badges are about how you thought - not how many answers were perfect. Circle a badge whenever you used that thinking move.

MISSION	CLUE FINDER	PATH BUILDER	BRAVE TRY	CHECK-IT	TRY AGAIN
1	☐	☐	☐	☐	☐
2	☐	☐	☐	☐	☐
3	☐	☐	☐	☐	☐
4	☐	☐	☐	☐	☐
5	☐	☐	☐	☐	☐

CASE DEBRIEF

Which thinking move felt strongest today?

Which mission made you say 'Hmm...' the longest?

Did you ever change your mind or try another path? What happened?

What was your favorite story moment?



How to Use This Prototype

The purpose is not to race through five correct answers.

This case practices what a child can do when the route is not immediately obvious: understand, find a foothold, build a path, try it, check it, and revise if needed. Story progression is the main reward; badges recognize productive thinking behaviors.

HOW TO FACILITATE

- 1 Let the child attempt each mission before giving procedural help.
- 2 Use the printed Hint Station one hint at a time.
- 3 Praise specific thinking moves: finding an anchor, drawing a model, testing, checking, or revising.
- 4 If an answer is wrong but the path is thoughtful, ask what the result teaches.
- 5 Do not require every badge. The tally is reflection, not a grade.
- 6 Watch whether story curiosity helps engagement or causes rushing.

MISSION MAP

#	PRIMARY DIMENSION	SECONDARY DIMENSIONS
1	Representation & Model Building	Anchor Finding; Checking
2	Working Backward	Anchor Finding; Decomposition
3	Decomposition	Anchor Finding; Representation
4	Elimination & Constraint Reasoning	Relationship Reasoning; Checking
5	Sequencing & Planning	Constraint Reasoning; Checking

Answers, Valid Paths, and What to Notice

1 - Which Way Is Up?

Answer: Turn the map 180 degrees. Valid approaches include anchoring on any landmark or experimentally rotating the page and checking all three.

Notice: Does the child manipulate the representation and verify with more than one landmark?

2 - The Clock That Stopped

Answer: 3:35. Example: 4:10 to 4:00 uses 10 minutes, then 25 more minutes backward. Another valid anchor is 3:40 after going back 30 minutes.

Notice: Can the child reason backward and find a friendly time anchor?

3 - The Three-Part Trail

Answer: 3 markers. Known parts total $3 + 4 + 2 = 9$; the whole is 12, leaving 3. Counting from 9 to 12 is also valid.

Notice: Can the child represent the whole as connected pieces and identify the unknown part?

4 - The Suspicious Shed

Answer: Benny Beaver carried the map. Pigeon = feather, Benny = boot print, Dottie = sandwich.

Notice: Does the child start from certainty, cross out impossibilities, and check every clue?

5 - Beat the Sunset

Answer: Both plans work. Plan A = 35 minutes and finishes at 5:55. Plan B = 36 minutes and finishes at 5:56.

Notice: Does the child include every action, compare with 40 minutes, and accept two valid plans?

PROTOTYPE QUESTION: After the case, write down one thing the story structure made better and one thing it made worse. Revise the next case from observed behavior.